

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

The radius (r), length (l) and resistance (R) of a metal wire was measured in the laboratory as

$$r = (0.35 \pm 0.05)\text{cm}$$

$$R = (100 \pm 10)\text{ohm}$$

$$l = (15 \pm 0.2)\text{cm}$$

The percentage error in resistivity of the material of the wire is :

(1) 25.6%

(2) 39.9%

(3) 37.3%

(4) 35.6%

Q2 - 2024 (01 Feb Shift 2)

Match List - I with List - II.

List - I

List - II

(Number)

(Significant figure)

(A) 1001

(I) 3

(B) 010.1

(II) 4

(C) 100.100

(III) 5

(D) 0.0010010

(IV) 6

Choose the correct answer from the options given below:

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(4) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Q3 - 2024 (29 Jan Shift 1)

The resistance $R = \frac{V}{I}$ where $V = (200 \pm 5)\text{V}$ and $I = (20 \pm 0.2)\text{A}$, the percentage error in the measurement of R is :

(1) 3.5%

(2) 7%

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(3) 3%

(4) 5.5%

Q4 - 2024 (29 Jan Shift 2)

A physical quantity Q is found to depend on quantities a, b, c by the relation $Q = \frac{a^4 b^3}{c^2}$. The percentage error in a, b and c are 3%, 4% and 5% respectively. Then, the percentage error in Q is :

(1) 66%

(2) 43%

(3) 34%

(4) 14%

Q5 - 2024 (30 Jan Shift 2)

A vector has magnitude same as that of $\vec{A} = 3\hat{j} + 4\hat{j}$ and is parallel to $\vec{B} = 4\hat{i} + 3\hat{j}$. The x and y components of this vector in first quadrant are x and 3 respectively where $x = \underline{\hspace{2cm}}$

Q6 - 2024 (31 Jan Shift 1)

If the percentage errors in measuring the length and the diameter of a wire are 0.1% each. The percentage error in measuring its resistance will be:

(1) 0.2%

(2) 0.3%

(3) 0.1%

(4) 0.144%

Q7 - 2024 (31 Jan Shift 2)

If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ , then

(1) $|\vec{A} - \vec{B}| = \sqrt{2}R \sin\left(\frac{\theta}{2}\right)$

(2) $|\vec{A} + \vec{B}| = 2R \sin\left(\frac{\theta}{2}\right)$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(3) $|\vec{A} + \vec{B}| = 2R \cos\left(\frac{\theta}{2}\right)$

(4) $|\vec{A} - \vec{B}| = 2R \cos\left(\frac{\theta}{2}\right)$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Answer Key

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Q1 (2) mathongo /// ma**Q2** (3) mathongo **Q3** (1) mathongo /// mc**Q4** (3) mathongo

Q5 (4) mathongo /// ma**Q6** (2) mathongo **Q7** (3) mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

Q1

$$\rho = R \frac{\rho}{\ell}$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta R}{R} + 2 \frac{\Delta r}{r} + \frac{\Delta \ell}{\ell}$$

$$= \frac{10}{100} + 2 \times \frac{0.05}{0.35} + \frac{0.2}{15}$$

$$= \frac{1}{10} + \frac{2}{7} + \frac{1}{75}$$

$$\frac{\Delta \rho}{\rho} = 39.9\%$$

Q2

Theoretical

Q3

$$R = \frac{V}{I}$$

According to error analysis

$$\frac{dR}{R} = \frac{dV}{V} + \frac{dI}{I}$$

$$\frac{dR}{R} = \frac{5}{200} + \frac{0.2}{20}$$

$$\frac{dR}{R} = \frac{7}{200}$$

$$\% \text{ error } \frac{dR}{R} \times 100 = \frac{7}{200} \times 100 = 3.5\%$$

Q4

$$Q = \frac{a^4 b^3}{c^2}$$

$$\frac{\Delta Q}{Q} = 4 \frac{\Delta a}{a} + 3 \frac{\Delta b}{b} + 2 \frac{\Delta c}{c}$$

$$\frac{\Delta Q}{Q} \times 100 = 4 \left(\frac{\Delta a}{a} \times 100 \right) + 3 \left(\frac{\Delta b}{b} \times 100 \right) + 2 \left(\frac{\Delta c}{c} \times 100 \right)$$

$$\% \text{ error in } Q = 4 \times 3\% + 3 \times 4\% + 2 \times 5\%$$

$$= 12\% + 12\% + 10\%$$

$$= 34\%$$

Q5

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$\frac{\vec{N}}{N} = \frac{|\vec{A}|\vec{B}}{5} = \frac{5(4\hat{i} + 3\hat{j})}{5} = 4\hat{i} + 3\hat{j}$$

$$\therefore x = 4$$

Q6

$$R = \frac{\rho L}{\pi \frac{d^2}{4}}$$

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + \frac{2\Delta d}{d}$$

$$\frac{\Delta L}{L} = 0.1\% \text{ and } \frac{\Delta d}{d} = 0.1\%$$

$$\frac{\Delta R}{R} = 0.3\%$$

Q7

The magnitude of resultant vector

$$R' = \sqrt{a^2 + b^2 + 2ab \cos \theta}$$

Here $a = b = R$

$$\text{Then } R' = \sqrt{R^2 + R^2 + 2R^2 \cos \theta}$$

$$= R\sqrt{2}\sqrt{1 + \cos \theta}$$

$$= \sqrt{2}R\sqrt{2\cos^2 \frac{\theta}{2}}$$

$$= 2R\cos \frac{\theta}{2}$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)